

NEW ULM, MN, USA

WMO: 726567

Lat: 44.319N

Lon: 94.502W

Elev: 1011

StdP: 14.17

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 04980

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.8	-8.6	-20.0	1.9	-12.5	-16.5	2.3	-8.7	33.7	12.5	30.7	15.8	8.0	280	0.588

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.1	90.2	74.1	87.6	73.0	83.9	71.0	77.9	85.7	75.7	83.4	73.8	80.8	12.7	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
75.2	137.3	82.7	73.0	127.0	80.7	71.9	122.4	79.4	42.2	85.6	39.9	83.5	38.0	79.7	91.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 27.3	24.2	20.6	DB	-18.7	95.7	5.9	4.1	-23.0	98.7	-26.4	101.1	-29.7	103.5	-34.0	106.5	(4)
(5)			WB	-19.7	81.7	6.4	3.2	-24.3	84.0	-28.0	85.8	-31.6	87.6	-36.2	89.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	45.6	14.8	19.2	31.8	46.1	58.3	68.5	72.2	69.3	62.3	48.8	34.1	20.2		
	DBStd	22.54	13.42	13.04	12.95	10.20	8.65	6.63	5.67	5.54	8.40	9.59	11.37	12.24		
	HDD50	4305	1092	863	578	189	24	0	0	0	7	140	488	924		
	HDD65	7728	1557	1283	1031	570	244	38	9	21	148	511	927	1389		
	CDD50	2698	0	0	13	71	281	556	688	599	377	102	11	0		
	CDD65	646	0	0	1	3	36	145	231	155	68	7	0	0		
	CDH74	5811	0	0	5	59	416	1291	2090	1199	657	92	2	0		
	CDH80	1770	0	0	0	15	131	423	676	314	189	22	0	0		
	WSAvg	9.3	10.4	10.4	10.2	10.7	10.2	8.5	7.2	6.6	8.2	9.7	10.0	9.8		
	PrecAvg	29.5	0.7	0.7	1.9	2.8	3.6	4.4	4.2	3.7	3.1	2.1	1.6	0.8		
(15) Precipitation	PrecMax	43.4	2.4	2.8	5.2	7.2	8.5	10.0	12.6	10.4	7.1	6.7	4.9	2.6		
	PrecMin	16.8	0.0	0.0	0.2	0.4	0.7	0.9	0.4	1.0	0.3	0.2	0.0	0.0		
	PrecStd	6.4	0.5	0.6	1.2	1.5	1.9	1.9	2.4	1.8	1.9	1.7	1.2	0.6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.3	54.2	71.5	81.4	90.0	93.0	93.3	90.6	90.2	82.4	69.6	52.3		
		MCWB	38.9	47.0	57.6	60.6	65.3	72.5	78.2	77.0	70.9	63.3	54.8	47.9		
	2%	DB	37.4	45.4	62.8	73.0	83.5	88.5	90.2	87.0	84.3	75.0	61.3	44.9		
		MCWB	34.4	41.3	54.4	55.7	64.4	71.2	77.1	75.3	69.5	60.9	51.2	39.9		
	5%	DB	35.8	39.3	55.1	68.1	79.0	84.2	87.5	82.8	81.5	69.9	55.0	38.7		
		MCWB	33.7	36.1	49.0	53.7	61.8	69.1	75.8	72.5	67.6	58.0	47.1	35.6		
	10%	DB	32.8	36.4	48.4	63.1	73.2	81.5	83.6	81.0	77.4	64.0	50.0	36.0		
		MCWB	31.4	34.3	44.4	50.8	59.3	67.2	73.1	70.5	65.5	55.2	44.0	33.9		
	0.4%	WB	39.4	47.9	61.8	62.8	71.4	77.2	82.2	79.9	75.6	69.2	58.3	48.3		
		MCDB	45.2	52.9	65.2	76.1	82.8	87.4	89.8	87.1	84.0	77.0	65.9	51.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	35.4	41.2	54.9	59.0	67.5	74.6	79.0	76.9	73.1	63.5	53.0	39.7		
		MCDB	37.8	45.5	60.4	69.2	76.9	83.5	86.5	83.9	80.4	69.3	60.1	43.6		
	5%	WB	33.4	37.2	50.0	55.5	64.6	72.4	76.9	74.6	70.6	60.0	48.3	36.3		
		MCDB	35.2	39.4	55.5	65.2	73.3	81.2	84.6	81.1	78.3	67.6	54.4	38.8		
	10%	WB	31.1	34.3	44.7	52.6	61.9	70.0	74.7	72.5	67.8	56.2	44.5	33.7		
		MCDB	32.5	36.2	49.0	61.2	70.3	78.2	81.7	78.6	75.0	63.7	49.3	35.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	15.9	16.5	16.1	19.2	20.0	19.1	19.1	19.1	21.9	19.9	16.8	14.8		
		MCDBR	17.3	18.3	24.9	28.8	27.8	23.9	21.4	21.2	25.6	27.7	24.9	19.2		
		MCWBR	14.6	14.7	16.8	15.6	13.1	11.2	11.1	11.4	12.6	14.8	16.0	15.5		
		MCDBR	15.7	17.0	23.1	23.9	22.0	20.2	19.0	19.1	21.7	22.4	22.4	17.5		
	5% WB	MCWBR	13.6	13.8	16.5	14.8	12.4	11.6	11.2	11.6	12.5	14.1	15.8	14.5		
(40) Clear-Sky Solar Irradiance	taub		0.260	0.284	0.306	0.362	0.384	0.401	0.411	0.411	0.370	0.322	0.284	0.264		
	taud		2.472	2.404	2.463	2.344	2.345	2.345	2.305	2.287	2.373	2.509	2.534	2.471		
	Ebn at Noon		281	290	297	286	281	276	271	267	270	272	269	266		
	Edh at Noon		22	28	30	37	39	39	40	39	33	25	21	20		
(44) All-Sky Solar Radiation	RadAvg		577	905	1189	1467	1710	1905	2052	1697	1344	881	572	430		
	RadStd		64	105	120	147	132	154	77	129	96	115	53	43		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	-1.71	-2.74	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (69 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page